

Week 7

Principal Component Analysis

Principal Component Analysis

Step 1: Compute the mean vector $\bar{X}$.

$$\bar{X} = \sum_{i=1}^n \frac{1}{n} x_i$$

Step 2: Compute the covariance matrix C

$$C = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})^T$$

Step 3: Find eigen values and eigen vectors.

Step 4: Choose eigen vectors corresponding to top k eigen values and derive the transformed data points.

$$\tilde{x}_i = \sum_{j=1}^k \alpha_j u_j, \text{ where } \alpha_j = (x_i^T u_j)$$

Step 5: Calculate the reconstruction error and projected variance as

$$\text{Reconstruction Error} = J = \frac{1}{n} \sum_{i=1}^n \|x_i - \tilde{x}_i\|^2$$

$$\text{Projected Variance} = \lambda_1 + \lambda_2 + \dots + \lambda_k$$

$$* \text{ data} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

$$* \text{ mean vector} = \bar{x} = \frac{1}{3} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

* Covariance Matrix

$$C = \frac{1}{3} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \end{bmatrix} \begin{bmatrix} -1 & -1 \end{bmatrix} \right)$$

$$= \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

* find Eigenvalues of matrix $C \rightarrow \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{bmatrix}$

$$\lambda^2 - \frac{4}{3}\lambda + \frac{2}{3} = 0 \Rightarrow \lambda_1 = 1, \lambda_2 = \frac{1}{3}$$

* find Eigenvector of $(A - \lambda I)$

$$\left. \begin{array}{l} \lambda_2 = 1 \\ \begin{bmatrix} \frac{2}{3} - 1 & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} - 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \end{array} \right| \begin{array}{l} \lambda_1 = 1/3 \\ \begin{bmatrix} \frac{2}{3} - \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} - \frac{1}{3} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \end{array}$$

Normalise Eigenvector

$$u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad | \quad u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

We want to reduce dimension to 1. So we take eigenvector corresponding to the top 1 eigenvalue and that is u_1

* Transforming data points:-

$$\tilde{x}_i = \sum_{j=1}^k \alpha_j u_j \text{ where } \alpha_j = (x_i^T u_j) \text{ and } k = \text{required dimension} \\ (1 \text{ in this case.})$$

$$\text{here } k=1$$

$$\tilde{x}_1 = \sum_{j=1}^1 \alpha_1 u_1 \quad \alpha_1 = x_1^T u_1.$$

$$= \alpha_1 \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\therefore \alpha_1 = x_1^T u_1 = \begin{bmatrix} 1 & 0 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}}$$

$$\therefore = \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$$

$$\tilde{x}_2 = \sum_{j=1}^1 \alpha_1 u_1 \quad \alpha_1 = x_2^T u_1.$$

$$= \alpha_1 \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\therefore \alpha_1 = x_2^T u_1 = \begin{bmatrix} 0 & 1 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}}$$

$$\therefore = \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$$

$$\tilde{x}_3 = \sum_{j=1}^1 \alpha_1 u_1 \quad \alpha_1 = x_3^T u_1.$$

$$= \alpha_1 \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\therefore \alpha_1 = x_3^T u_1 = \begin{bmatrix} -1 & -1 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = -\sqrt{2}$$

$$\therefore = -\sqrt{2} \times \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

* Reconstruction Error.

Original Data	Projected Data	Error
$x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\tilde{x}_1 = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} 1/2 \\ -1/2 \end{bmatrix}$
$x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$	$\tilde{x}_2 = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 1/2 \end{bmatrix}$
$x_3 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$	$\tilde{x}_3 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$	$\begin{bmatrix} -1 \\ -1 \end{bmatrix} - \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$\text{Reconstruction Error} = J = \frac{1}{n} \sum_{i=1}^n \|x_i - \tilde{x}_i\|^2,$$

$$J = \frac{1}{3} \left(\left\| \begin{bmatrix} 1/2 \\ -1/2 \end{bmatrix} \right\|^2 + \left\| \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix} \right\|^2 + \left\| \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\|^2 \right)$$

$$= \frac{1}{3} \left(\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} \right) = \boxed{\frac{1}{3}} \text{ Ans.}$$